

Online Appendix

The Upside of Being Replaceable: AI, Market Structure, and Growth

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[Main paper](#) | [Code and data](#)

S1 Additional competitive-to-monopoly simulation results

I supplement the $\sigma = 1.5$ comparison in Section 7.4 of the main paper with the four-elasticity exercises at both research productivities: $\sigma = 0.9, 1.0, 1.1, 1.5$ and $\chi = 7.5, 1.5$. Each economy starts on a competitive labor-bottleneck balanced growth path. At date zero, an unexpected permanent grant of exclusive AI rights introduces monopoly pricing and incentives for RSI. Continued competition provides the benchmark from the same initial stocks. I use the existing simulated trajectories; no additional equilibria are computed for this presentation. Appendix B.1 of the main paper documents the initial conditions and numerical checks.

The level figures retain the three windows used in that section; the growth, price, and distributional figures below show two pre-event years and years 0–10 above, with years 10–500 below. Thin horizontal lines show continued competition; black dotted long-run lines show the analytical monopoly limits at $\sigma = 1.5$. The benchmark lines also distinguish the competitive income shares from the monopoly limits: competitive AI revenue pays for inference, whereas monopoly revenue also finances research and developer profit. The date-zero level changes are not finite growth rates: open and filled dots distinguish the values on either side of the event, and growth curves describe the smooth trajectory on each side separately.

Table S1: Competitive initial capital and the impact of exclusive AI rights

σ	K_0	Change in output	Change in wage
0.9	6.235	−32.4%	−36.6%
1.0	6.278	−17.6%	−17.6%
1.1	6.324	−13.7%	−11.9%
1.5	6.535	−10.4%	−5.4%

Notes: The capital stocks are derived from the competitive BGP, not selected independently. All other parameters and initial stocks, except capital, are in Table 2 of the main paper. Changes compare levels immediately after and before the grant. They are identical for both values of χ , since K and B cannot jump. The ratio $K_0/Y_{0-} = 3.3$ applies before the grant, not after output falls.

Level comparisons across elasticities. Figures S1 and S2 preserve the level comparisons for all four elasticities. Research can produce level gains even where it does not permanently raise growth. With $\chi = 7.5$, output first regains its competitive counterpart around year 18.2 at $\sigma = 1.1$ and year 46.9 at $\sigma = 1$. With $\chi = 1.5$, those recoveries occur around years 87.3 and 212.3. At $\sigma = 0.9$, neither output nor wages regain their competitive counterparts within 500 years under either χ . The initial AI-service price in that case is about 131 times its competitive level, a large distortion implied by the illustrative parameters rather than an empirical estimate. These recovery dates compare contemporaneous levels; they do not rank welfare.

Higher research productivity. Figures S3–S5 report $\chi = 7.5$. Unlike the pure RSI-activation experiment, the grant immediately reduces the return on capital: output falls while capital cannot jump, so equation (11) in the main paper implies a lower interest rate. The Euler equation (5) then gives lower consumption growth immediately after the grant, even where the level of consumption jumps upward. Subsequent RSI and capital adjustment determine recovery. Research absorbs resources before its gains are fully realized, and the distribution panels separate developer net profit from expenditure on inference and research.

Lower research productivity. Figures S6–S8 report $\chi = 1.5$. The impact pricing distortion is unchanged, but the initial research allocation and efficiency growth are smaller. At $\sigma = 1.5$, the eventual rise in growth and decline in labor’s share emerge later. At year 500, growth and returns still exceed their monopoly limits, so the final displayed values should not be read as those limits.

Replication. The repository’s [competitive-transition guide](#) records the parameters, the institutional experiment, and the replication commands. The driver is `scripts/simulate_competitive_to_monopoly.py`; the renderer is `scripts/report_competitive_to_monopoly.py`. The guide links to the separate results folders for each research productivity, containing the original paths, competitive benchmarks, impact changes, and numerical audits. The main-text $\sigma = 1.5$ comparison is rendered by `scripts/report_competitive_main_comparison.py`, which reads these same audited paths without changing them. Figure windows only select dates from those stored paths; they do not change the simulated economy or the numerical solution horizon.

Levels relative to continued competition | $\chi = 7.5$

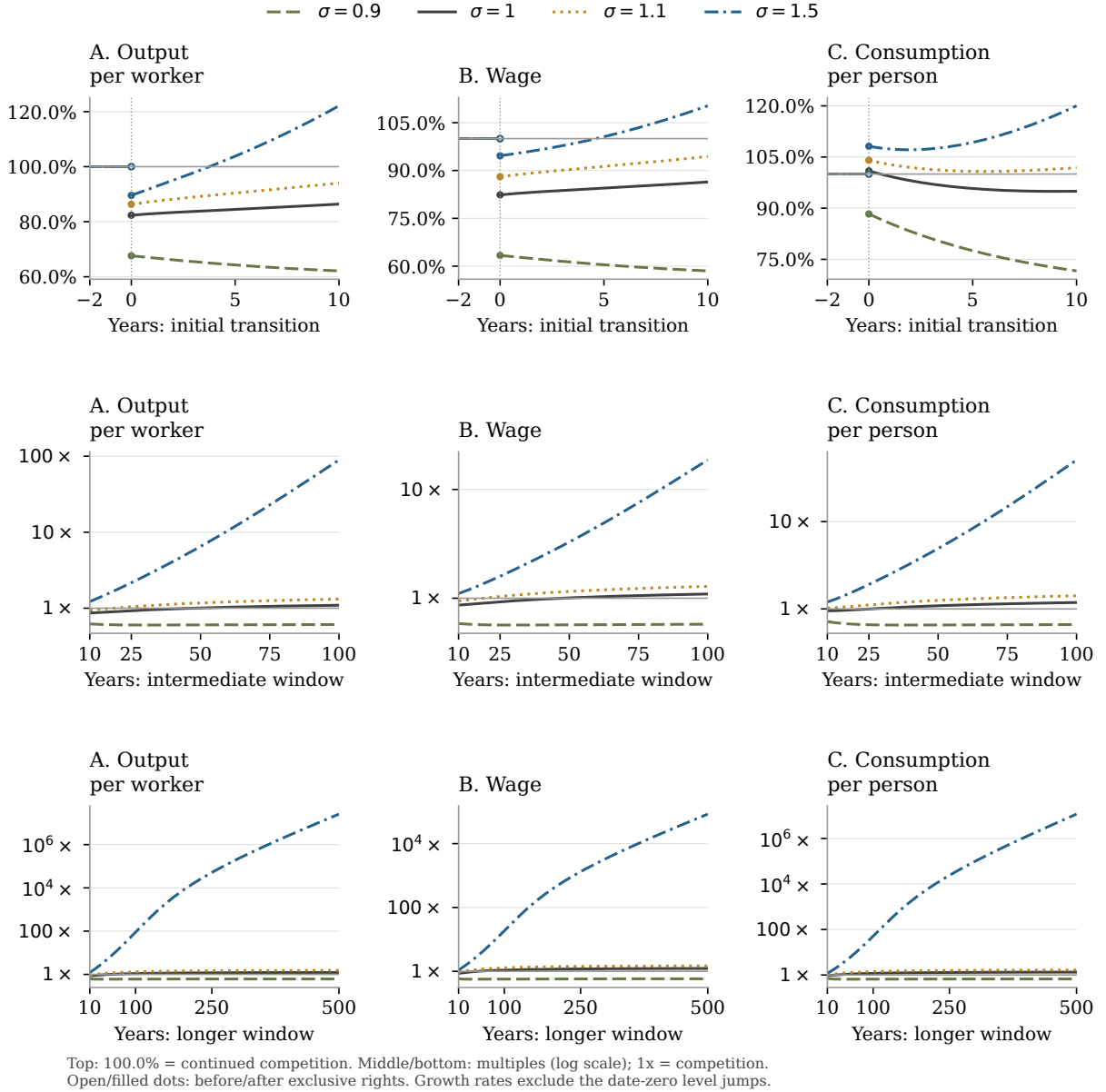
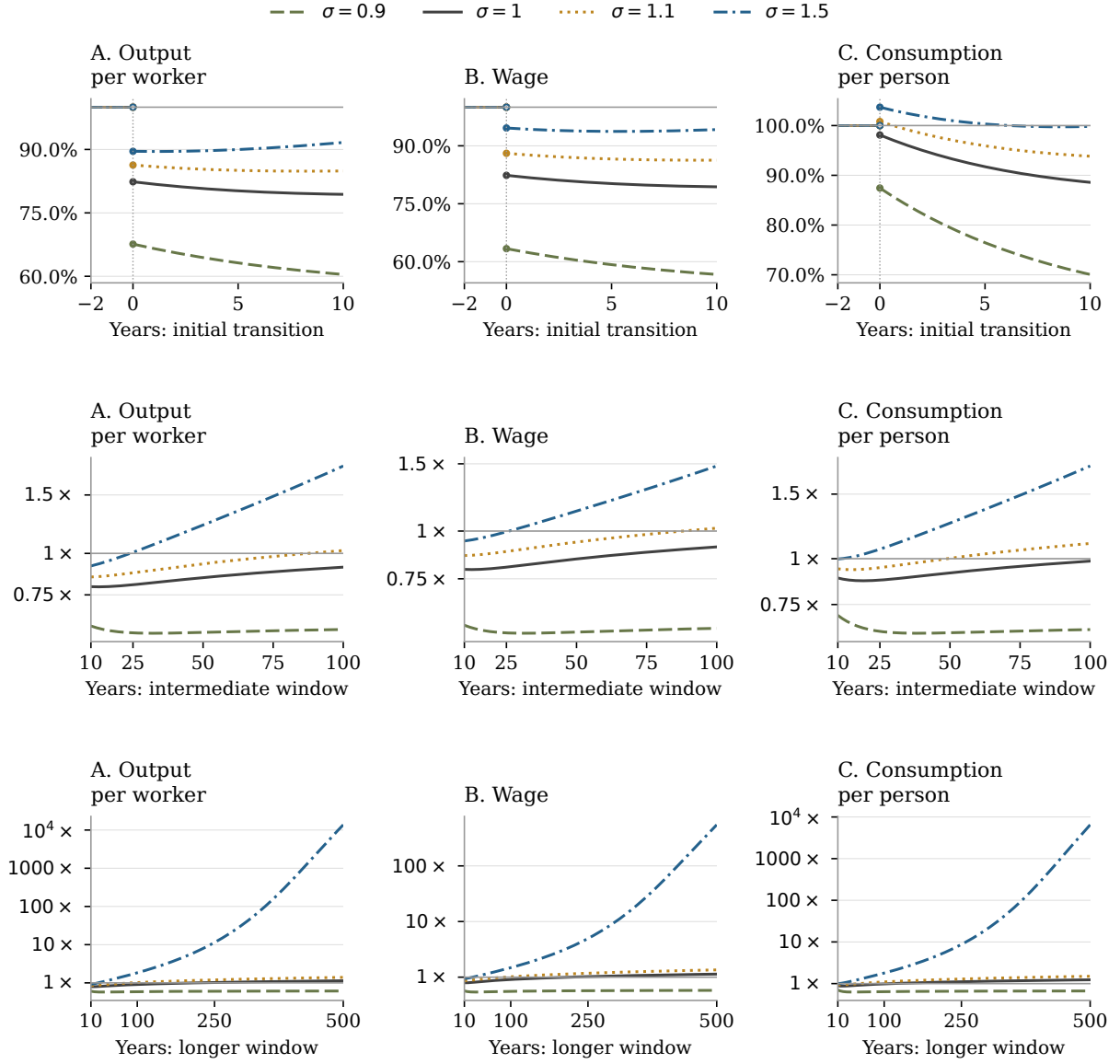


Figure S1: From competition to monopoly with RSI, $\chi = 7.5$: levels relative to continued competition for all four elasticities. The rows cover years $-2-10$, $10-100$, and $10-500$. The upper row uses percentages; the middle and lower rows use multiples on logarithmic scales. The horizontal benchmark is 100.0% or 1x. Open and filled dots mark levels before and after the grant. Vertical scales differ across windows and between the two research-productivity exercises.

Levels relative to continued competition | $\chi = 1.5$



Top: 100.0% = continued competition. Middle/bottom: multiples (log scale); 1x = competition.
Open/filled dots: before/after exclusive rights. Growth rates exclude the date-zero level jumps.

Figure S2: From competition to monopoly with RSI, $\chi = 1.5$: output per worker, wages, and consumption per person relative to continued competition for all four elasticities. Windows, normalizations, and line styles match Figure S1. Vertical scales are fitted to this exercise.

Output, AI services, and capital growth | $\chi = 7.5$

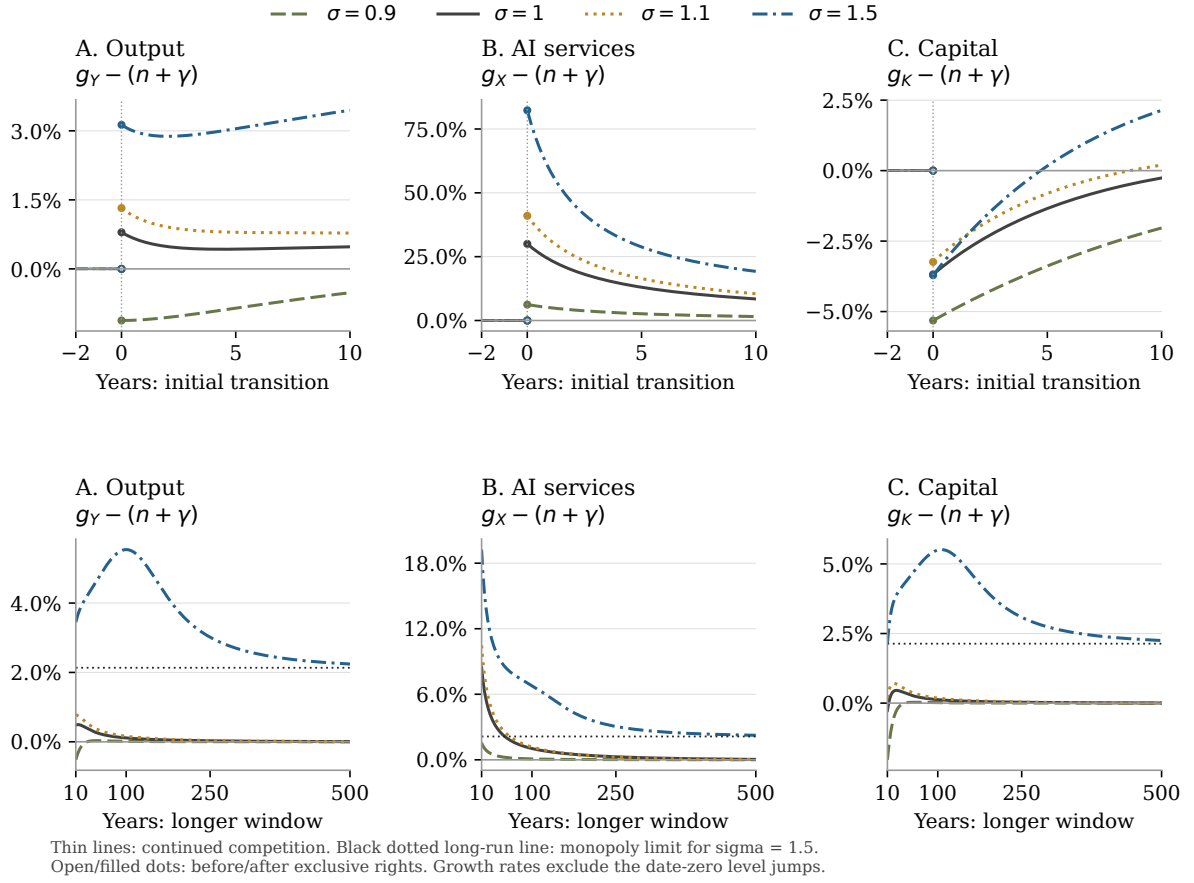


Figure S3: Competition to monopoly, $\chi = 7.5$: annual growth of output, AI services, and capital relative to effective labor. Zero denotes growth at $n + \gamma$. Growth rates exclude the discrete level changes at date zero.

Wages, interest rates, and AI prices | $\chi = 7.5$

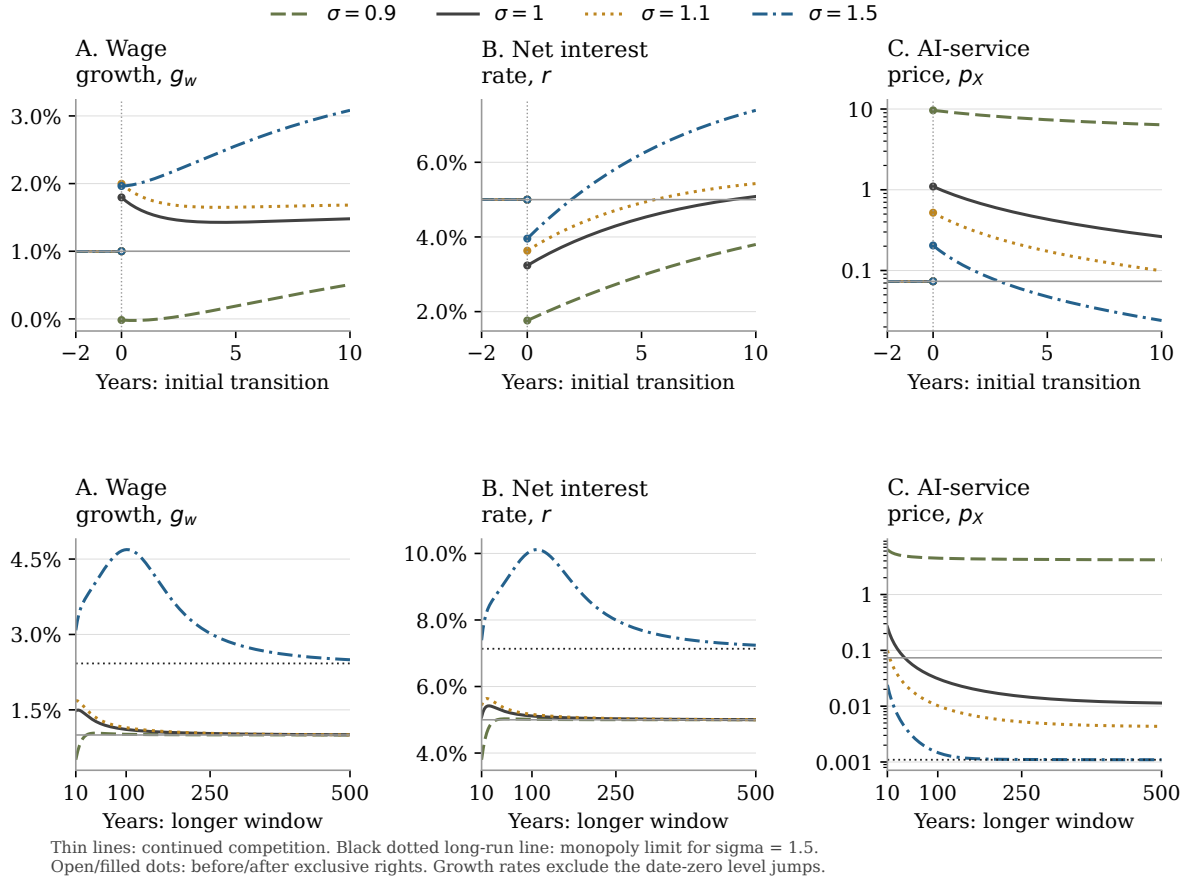
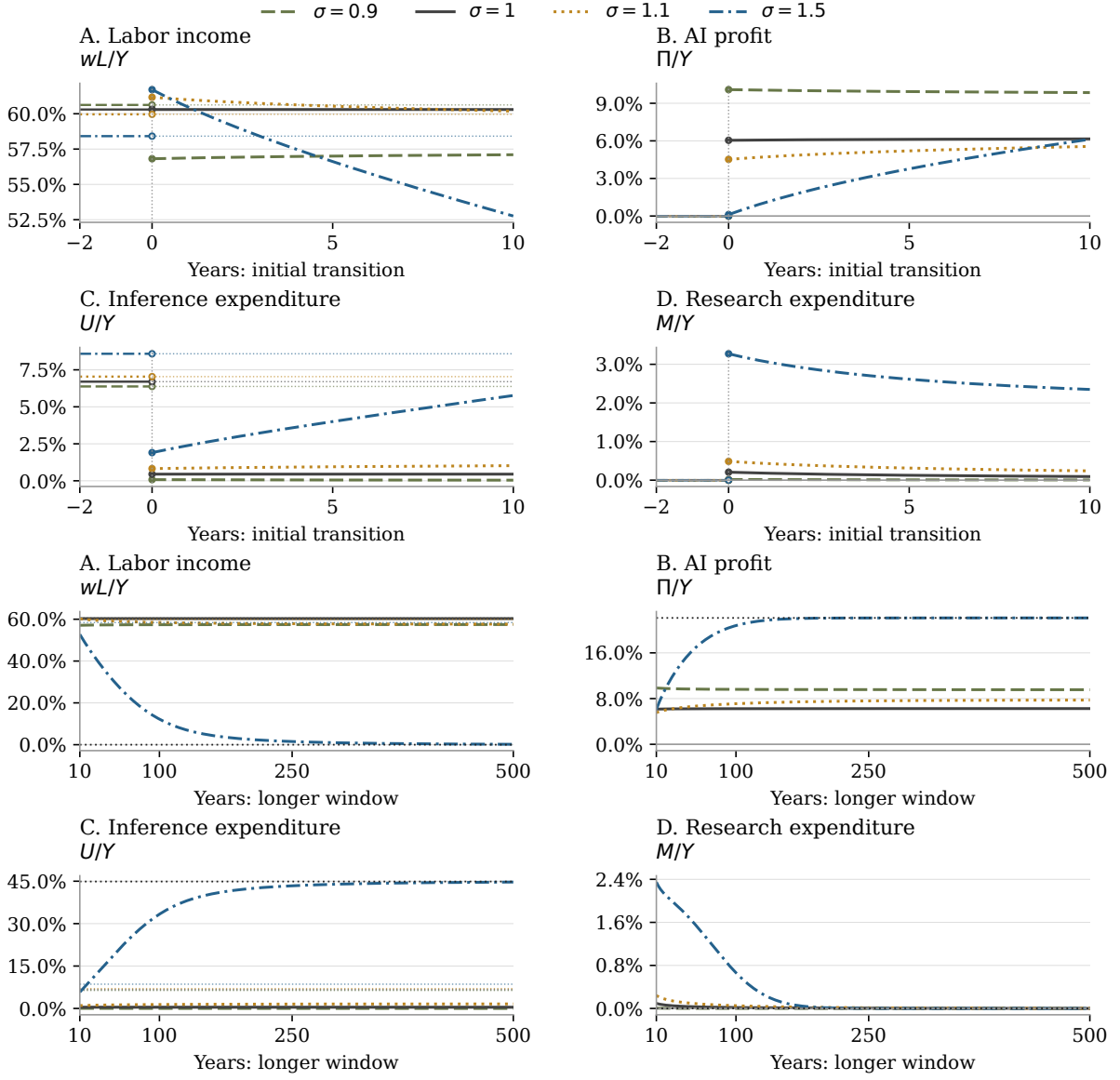


Figure S4: Competition to monopoly, $\chi = 7.5$: annual wage growth, the net interest rate, and AI-service prices. Prices are measured in final-good units on a logarithmic scale. The wage level falls at the grant; panel A reports wage growth on either side, not that level change.

Income and compute expenditure shares | $\chi = 7.5$



Thin lines: continued competition. Black dotted long-run line: monopoly limit for $\sigma = 1.5$.
Open/filled dots: before/after exclusive rights. Growth rates exclude the date-zero level jumps.

Figure S5: Competition to monopoly, $\chi = 7.5$: labor income, developer net profit, inference expenditure, and research expenditure as shares of output. The first two rows cover years $-2-10$; the last two cover years $10-500$. Gross capital income remains 33.0% of output.

Output, AI services, and capital growth | $\chi = 1.5$

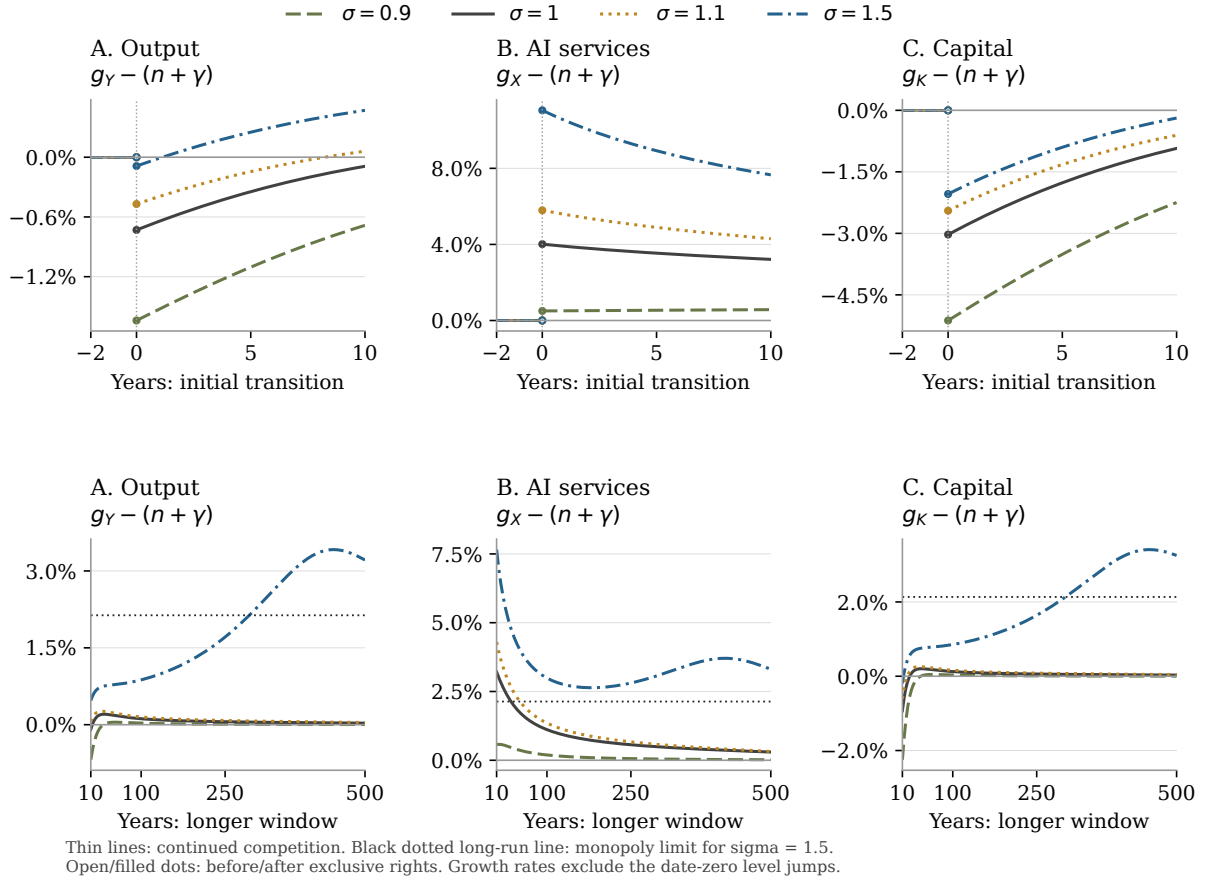


Figure S6: Competition to monopoly, $\chi = 1.5$: annual growth of output, AI services, and capital relative to effective labor. Windows and benchmarks match Figure S3.

Wages, interest rates, and AI prices | $\chi = 1.5$

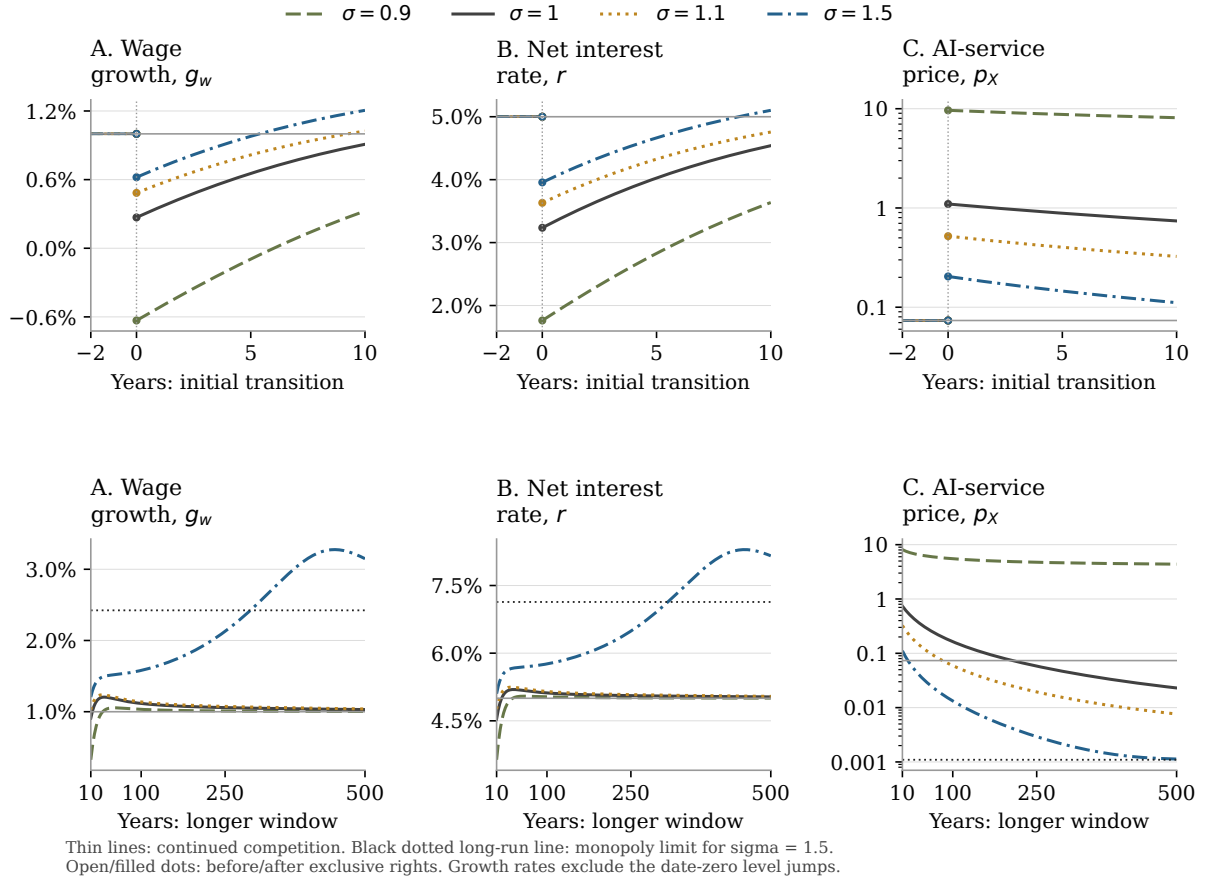
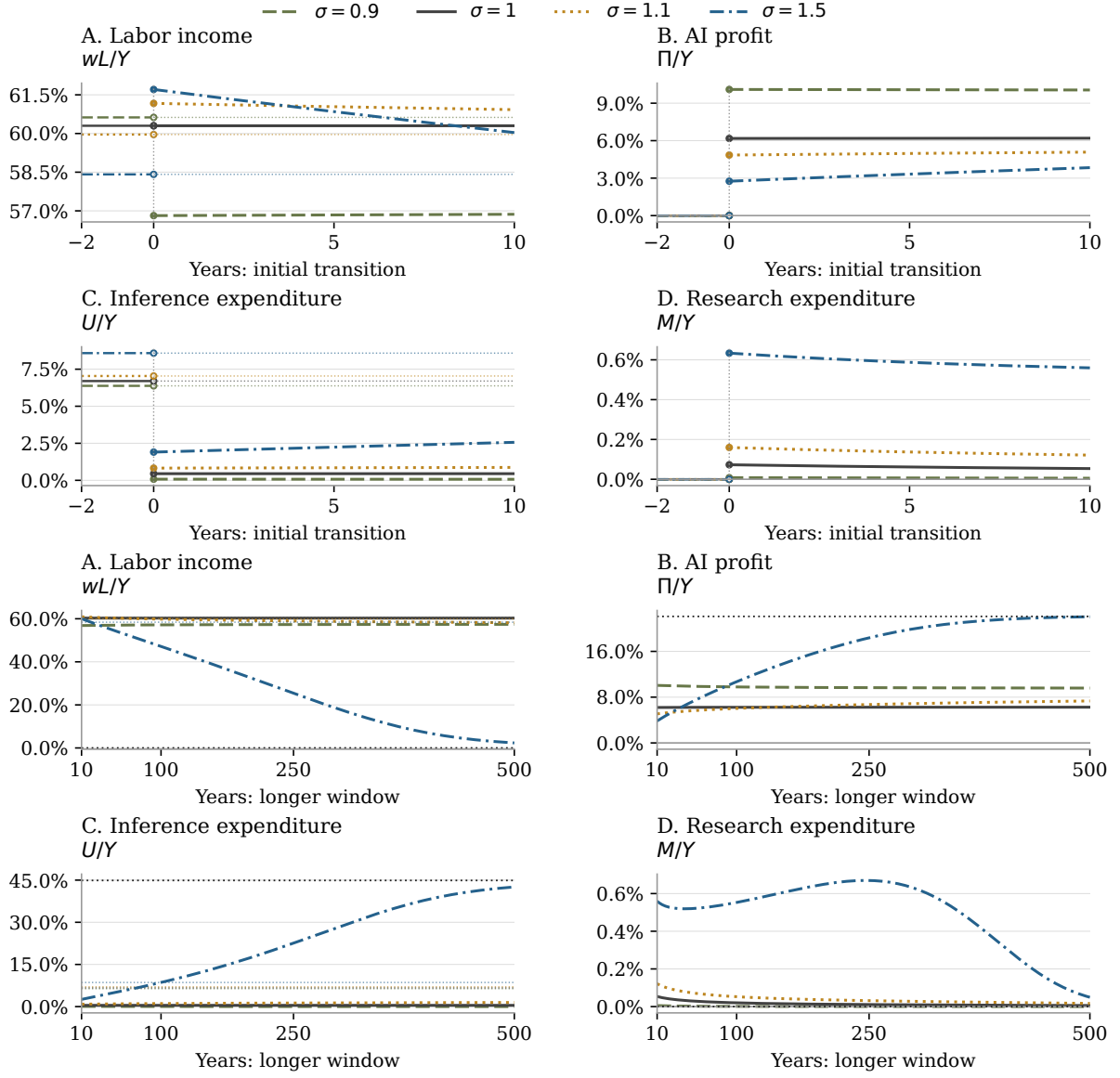


Figure S7: Competition to monopoly, $\chi = 1.5$: annual wage growth, the net interest rate, and AI-service prices in final-good units. Windows, benchmarks, and the logarithmic price scale match Figure S4.

Income and compute expenditure shares | $\chi = 1.5$



Thin lines: continued competition. Black dotted long-run line: monopoly limit for $\sigma = 1.5$.
 Open/filled dots: before/after exclusive rights. Growth rates exclude the date-zero level jumps.

Figure S8: Competition to monopoly, $\chi = 1.5$: labor income, developer net profit, inference expenditure, and research expenditure relative to output. Layout and benchmarks match Figure S5.

S2 Monopoly slows growth

I now ask whether monopoly can eliminate AI-dominated growth even while research makes AI more efficient. This is the opposite outcome to the exercise in Section 7.4 of the main paper: research gains fail to compensate for the restriction on AI use caused by monopoly pricing. Following Corollary 1 of the main paper, I choose $\sigma = 1.5$ and $\mathcal{B}^{\text{mc}}(1.5) < B_0 < \bar{B} < \mathcal{B}(1.5)$. Initial efficiency is high enough for AI-dominated growth under competition, but even the upper bound is insufficient under monopoly. As in the main-text experiment, the unexpected grant covers existing technology and future improvements. RSI is available throughout; the grant changes pricing and research incentives. The preceding simulations remain separate experiments with different initial states and efficiency bounds.

Table S2 reports the new design. I retain the other parameters in Table 2 of the main paper and compare continued competition with monopoly under $\chi = 7.5$ and $\chi = 1.5$, all from the same initial stocks. Before granting monopoly rights, I let the competitive economy approach its AI-dominated growth path. I select the first whole year when output-per-worker growth, wage growth, and the interest rate are each within 0.1 percentage point of their long-run limits, and reset the date to zero. I inherit A_0 , N_0 , and K_0 jointly from that date, keeping B_0 and both values of χ unchanged. This is a near-long-run competitive allocation, not an exact steady state: $K/(AL)$ continues to grow. Just before the grant, output per worker grows at 5.8%, wages at 4.2%, and the interest rate is 9.8%. The capital-output ratio is 2.23. Section S2.1 below describes the additional competitive-transition solver and numerical checks.

Table S2: Parameters and initial stocks for the growth-reversal experiment

Object	Value	Choice
σ	1.5	Substitution
χ	7.5; 1.5	Same illustrative research productivities
$\bar{B}$	1,113.5	$0.90\mathcal{B}(1.5)$
B_0	1,002.2	$0.90\bar{B}$
A_0, N_0	6.6859; 1.7683	Inherited from competitive prehistory
K_0	7.2907×10^9	Inherited from the same date
K_0/Y_{0-}	2.23	Competitive ratio before the grant

Notes: $\mathcal{B}^{\text{mc}}(1.5) = 829.0$ and $\mathcal{B}(1.5) = 1,237.3$. The upper bound and initial efficiency are chosen to satisfy the corollary's threshold inequalities. The three paths share the reported initial stocks; consumption and research are equilibrium choices. The initialization uses 190 years of competitive prehistory, not a forecast of a policy date. Stocks retain their original units; these illustrative choices are not empirical estimates. Section S2.1 below describes the prehistory and its starting stocks.

Figure S9 shows annual output-per-worker growth, wage growth, and the net interest rate in panels A–C. The rows cover years -10 – -10 , 10 – 100 , and 100 – 500 ; negative dates show the common competitive prehistory. The dashed gray line denotes continued competition; the solid blue and dash-dotted gold lines denote monopoly with high and low research productivity. At the grant, output falls by 55.6%, wages by 33.5%, and AI services by 70.3%. These are instantaneous level changes at unchanged K and B , not annual growth rates in the figure. The lower output-capital

ratio reduces the interest rate from 9.8% to 1.6%.

After this level loss, high research productivity briefly raises output-per-worker growth to 13.4% as efficiency improves rapidly. With low research productivity, initial growth is only 0.1%. Neither response restores AI-dominated growth: by year 10, output per worker contracts under both monopoly paths. The inherited capital stock supported faster growth under competitive AI supply; restricting that supply lowers its return and induces capital decumulation. Consumption initially jumps upward by 141.6% and 132.2%, respectively, as lower compute use and investment release resources despite the fall in gross output. Consumption then falls: the interest rate initially lies below ρ , so the Euler equation implies negative consumption-per-person growth after the grant.

The longer windows show that the loss of AI-dominated growth is not confined to the initial repricing. Under continued competition, output-per-worker growth tends to 5.7%, wage growth to 4.1%, and the interest rate to 9.7%, as in equation (46) of the main paper. Under either monopoly path, the corresponding limits are 1.0%, 1.0%, and 5.0%, consistent with the labor-bottleneck regime in Proposition 1 of the main paper. Both monopoly economies are still adjusting at year 500: output-per-worker growth is about -0.1% , wage growth is 0.3% , and the interest rate is 3.9% . The large stock of capital inherited from the competitive expansion takes a long time to adjust to restricted AI supply. The monopoly limit therefore remains distinct from the last plotted date: monopoly removes permanently faster growth, not all long-run growth.

Figure S10 explains why efficiency gains do not prevent this outcome. Using the same lines and windows, panels A–C show $B/\bar{B}$, AI services relative to continued competition, and labor income as a share of output. In the first decade, high research productivity takes efficiency to 99.9% of its upper bound, compared with 97.1% under low research productivity. Both nevertheless supply fewer AI services than continued competition. In the longer run, efficiency approaches $\bar{B}$, an improvement of 11.1% over B_0 ; competition retains B_0 . Yet the monopoly markup prevents these improvements from sustaining the joint expansion of capital and AI services that occurs under competition. AI services under monopoly become negligible *relative to the competitive path*, not in absolute terms.

Labor’s income share tends to 4.5% under monopoly rather than zero under competition. This larger limiting share coexists with slower wage growth. At year 500, the labor share is still only 2.1% and 2.2% under high and low research productivity, respectively, confirming that neither path has reached its labor-bottleneck limit.

Together with Section 7.4 of the main paper, this experiment illustrates opposite regime rankings. There, research allows the monopoly economy with $\sigma = 1.5$ to escape the labor bottleneck from a low initial efficiency. Here, initial efficiency already sustains AI-dominated growth under competition, and further efficiency gains do not compensate for the restriction on AI use caused by monopoly pricing. More efficient AI is therefore compatible with slower long-run growth.

The large initial adjustments also reflect the abrupt withdrawal of access to existing technology. I use this experiment to illustrate a possible growth reversal, not to predict the effects of an actual monopoly grant or rank household welfare.

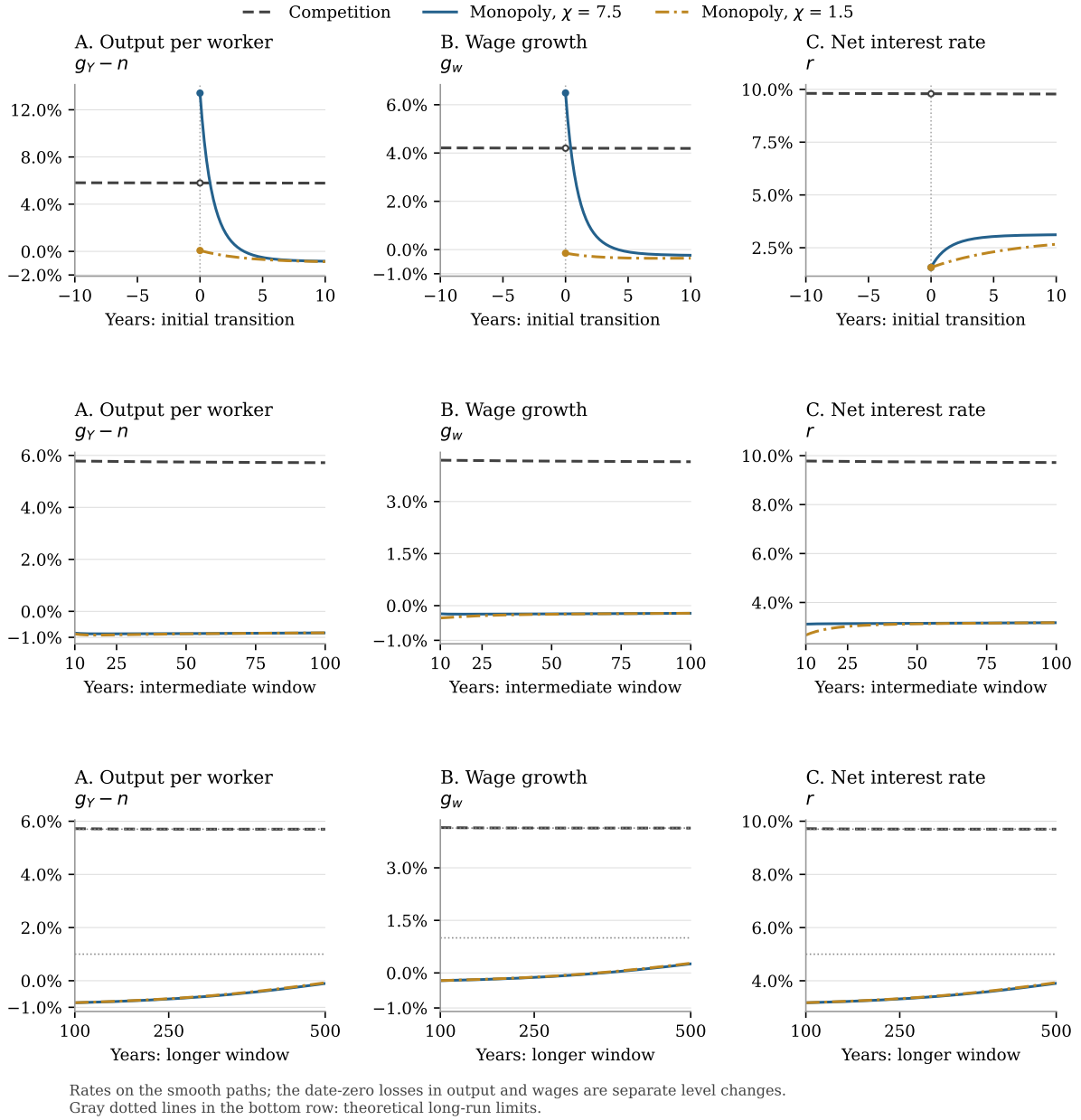
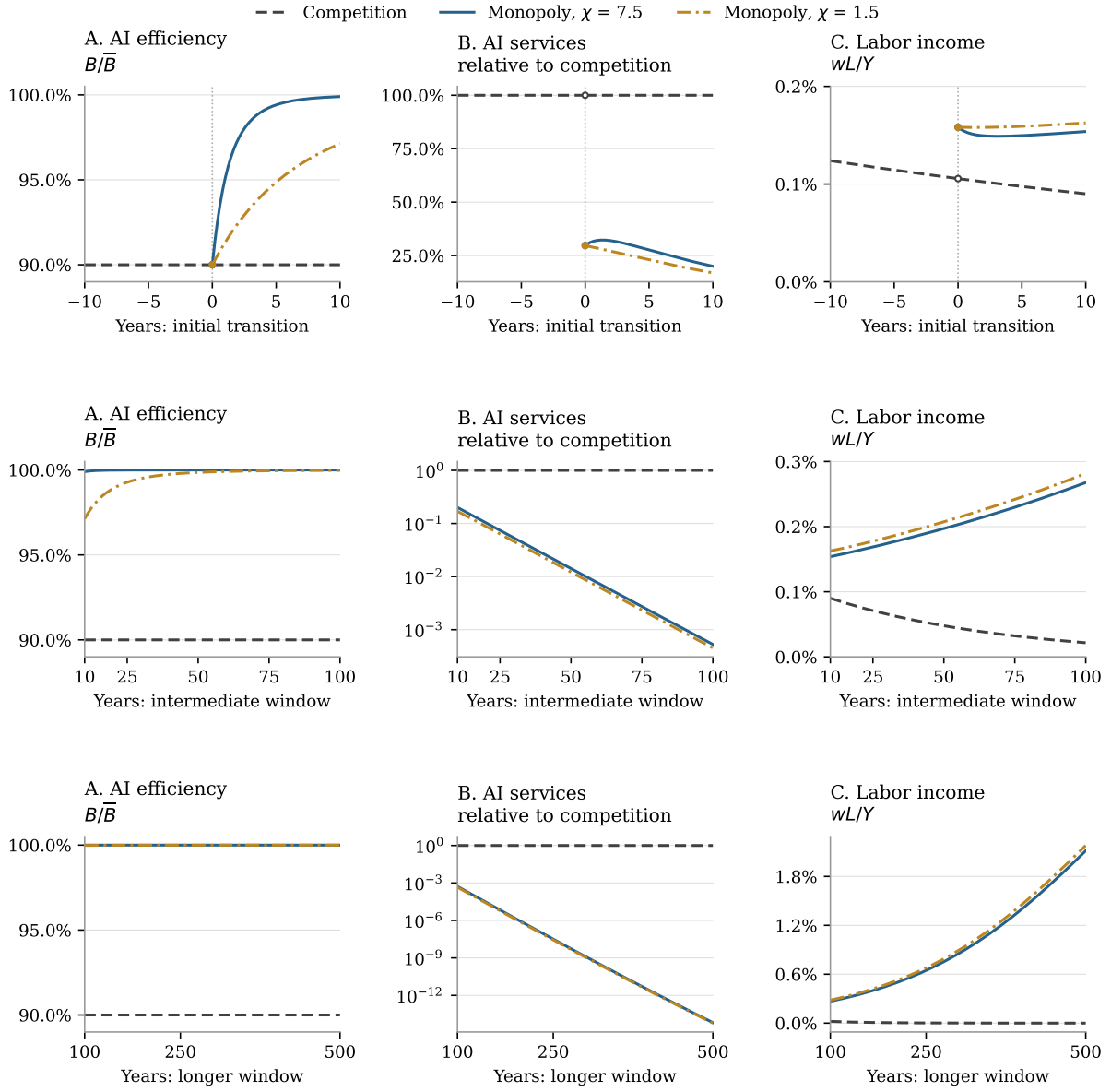


Figure S9: Monopoly and the loss of AI-dominated growth: annual output-per-worker growth, wage growth, and the net interest rate. Rows show years $-10-10$, $10-100$, and $100-500$. The grant occurs at zero, after the common competitive prehistory. Rates describe the smooth paths in each scenario, not the instantaneous output and wage losses. Gray dotted lines in the bottom row indicate theoretical long-run limits. Vertical scales differ across windows.



Panel B: 100.0% or 1 = continued competition; logarithmic scale in the middle and bottom rows. All paths share initial capital and AI efficiency. Vertical scales differ across windows.

Figure S10: More efficient AI but slower growth under monopoly: efficiency relative to its upper bound, AI services relative to continued competition, and labor's income share. Windows and line styles match Figure S9. Panel B uses percentages in the first row and multiples on logarithmic scales in the lower rows; its competitive benchmark is 100.0% or one. Vertical scales differ across windows.

S2.1 Numerical construction and checks

I construct the competitive prehistory with $B = B_0$ above the competitive threshold. At the start of this prehistory, I normalize $A = N = 1$ and set $K = 14,047.3$, the limiting monopoly capital-effective-labor ratio at the chosen upper bound. These seed values are illustrative; they are not the stocks at the monopoly grant. I advance the competitive path to year 190, the first whole year at which the three rates in (46) are each within 0.1 percentage point of their limits; all subsequent annual observations through year 1,000 also satisfy this criterion. I then set that date to zero and inherit A , N , and K without changing units. Re-solving the competitive equilibrium from those stocks reproduces the continued prehistory path to within 10^{-9} in log levels over the next 500 years. The figures include its last ten pre-grant years.

I solve the competitive resource constraint and Euler equation in (97), with $M = 0$ and $B = B_0$, using the inherited capital stock and the terminal condition $C/K \rightarrow \rho - n$ from (99). The numerical coordinates are $\log(C/K)$ and $\log(K/(AL)) - [\mathcal{R}^{\text{mc}}(B_0) - \rho - \gamma]t$. Removing the limiting trend avoids extremely large levels without imposing balanced growth on the transition. I use the `scipy.integrate.solve_bvp` routine described in Appendix B of the main paper. Extending its horizon from 1,200 to 1,800 years changes these coordinates by less than 10^{-9} over the displayed 500 years; independent finite-difference residuals are below 10^{-9} . The static supply condition, positive allocation, limiting growth rates, and household transversality diagnostics also pass the checks.

I solve the two monopoly paths using the continuation method and equilibrium checks in Appendix B of the main paper. Their final solution horizon is about 4,592 years, although the figures display only the first 500 years. Both pass the equation, horizon-extension, transversality, and developer-optimality checks, including denser checks over the first decade. The [replication guide for this experiment](#) records the inputs, data, audit reports, and commands.